

Generative AI

Developments since ChatGPT

Agenda

- Recap: Training LLMs
- Developments after ChatGPT was released
 - Prompting
 - In-context Learning
 - Reasoning
 - Retrieval Augmented Generation
 - Agentic AI
 - Political dimension of LLMs
- Midterm feedback (if time allows)

Resources

“Speech and Language Processing”, Aug. 2025. Jurafsky & Martin

7.3 Prompting

10.4 & 10.5 Test-time compute

11.1 & 11.3 Information Retrieval/Answering Questions with RAG

“Hands-on LLMs”. Alammam & Grootendorst

6 Prompt Engineering

Large language models are zero-shot reasoners. (Kojima et al., NeurIPS 2022)

Blog post on Reasoning

Blog on Agentic AI

Recap

The three stages of training

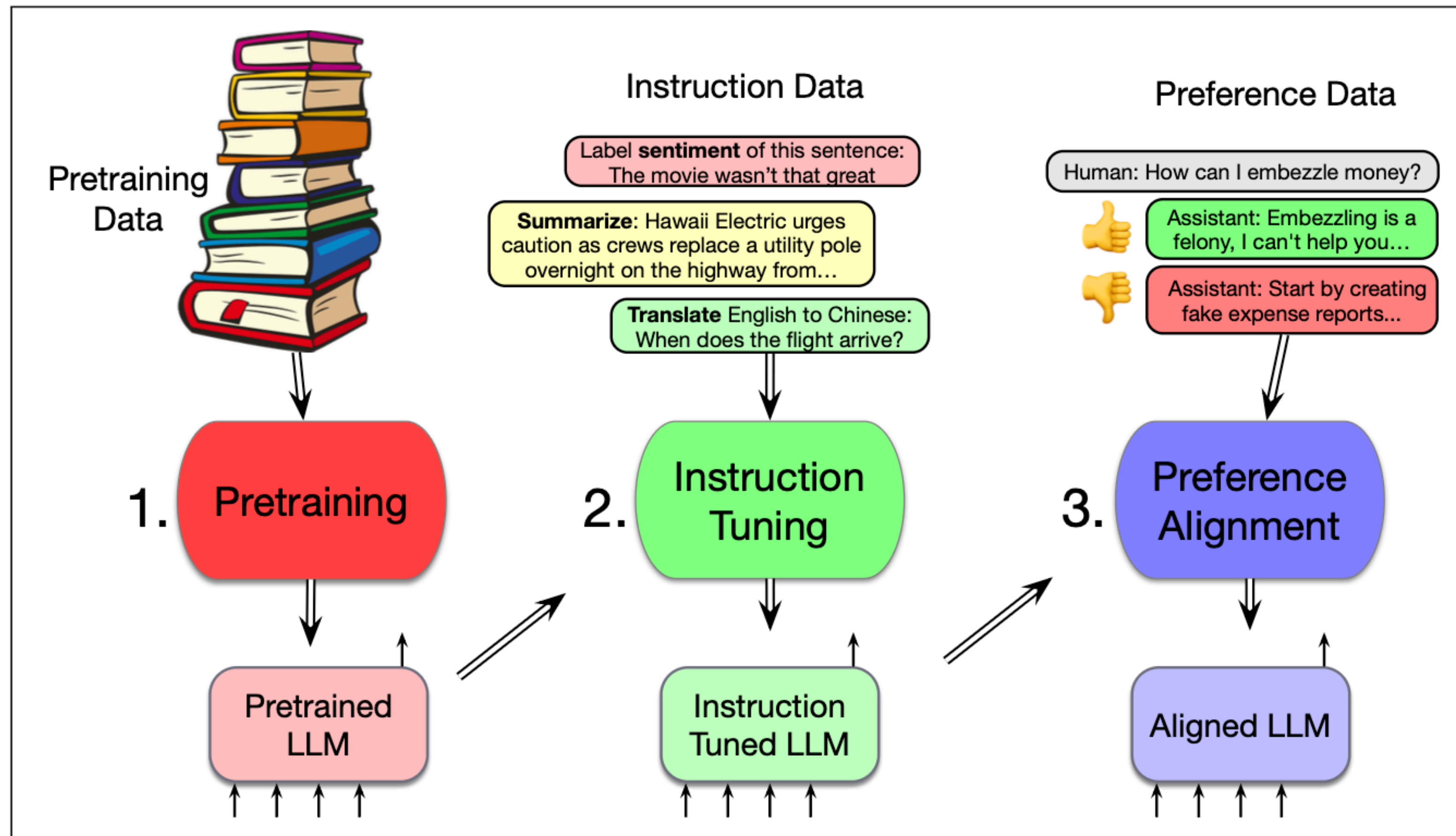


Figure 7.12 Three stages of training large language models: pretraining, instruction tuning, and preference alignment.

Generative AI

What is generative AI?

- “a subfield of AI that uses generative models to produce amongst others  text,  images,  videos,  audio” [Wikipedia]
- Large Language Models considered at the forefront of genAI
- popular LLMs:
 - open-weight: model weights are released and models can be locally explored **!!** usually not open-source **!!**
 - Llama, Gemma, Qwen
 - proprietary: models are closed-source and can only be accessed through APIs, usually monetized 
 - GPT, Gemini, Claude



Llama models

How a model evolves over time

	Training Data	Params	Context Length	Tokens	Modality	Languages	Release
Llama2	Mix of publicly available data	7B/13B/70B	4k	2 T	Text	en	07/23
Llama 3.1	Mix of publicly available data	8B/70B/405B	128k	15T +	Text	en, de, fr, hi, it, pt, es, th	07/24
Llama 4	Public data plus data from Meta	17B activated (109B/409B)	10M	40T	Input: Text + Image Output: Text	ar, en, de, fr, hi, id, it, pt, es, tl, th, vi	05/25

Prompting

Prompting

Prompts usually consist of 2 parts: a system prompt and a user prompt

- system prompt: *the how and why*
 - the first instruction to the LLM
 - defines the task or role for the LLM
 - sets tone and context
- user prompt: *the what*
 - the specific instruction
 - might change in every iteration, e.g. when data points change

Prompting

a minimal system prompt:

```
<system>You are a helpful and knowledgeable assistant. Answer  
concisely and correctly.
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```

system and user prompt are used to condition the generation of
subsequent text

Prompting

Another example from Claude (LLM developed by Anthropic)

Claude should give concise responses to very simple questions, but provide thorough responses to complex and open-ended questions.

Claude is able to explain difficult concepts or ideas clearly. It can also illustrate its explanations with examples, thought experiments, or metaphors.

Claude does not provide information that could be used to make chemical or biological or nuclear weapons

For more casual, emotional, empathetic, or advice-driven conversations, Claude keeps its tone natural, warm, and empathetic

Claude cares about people's well-being and avoids encouraging or facilitating self-destructive behavior

If Claude provides bullet points in its response, it should use markdown, and each bullet point should be at least 1-2 sentences long unless the human requests otherwise

Prompting

Another example from Claude (LLM developed by Anthropic)

format

Claude should give **concise** responses to very **simple** questions, but provide **thorough** responses to **complex and open-ended** questions.

Claude is able to explain difficult concepts or ideas clearly. It can also **illustrate its explanations** with examples, thought experiments, or metaphors.

safety

Claude does not provide information that could be used to make chemical or biological or nuclear weapons

tone

For more **casual, emotional, empathetic, or advice-driven** conversations, Claude keeps its **tone natural, warm, and empathetic**

Claude cares about people's well-being and avoids encouraging or facilitating self-destructive behavior

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Prompting Strategies

- Depending on the task and the desired outcome, a different prompting strategy is needed
- Instructions can include
 - length restrictions
 - personas to assume
 - requirements for a specific file format

Prompting Strategies

Simple completion

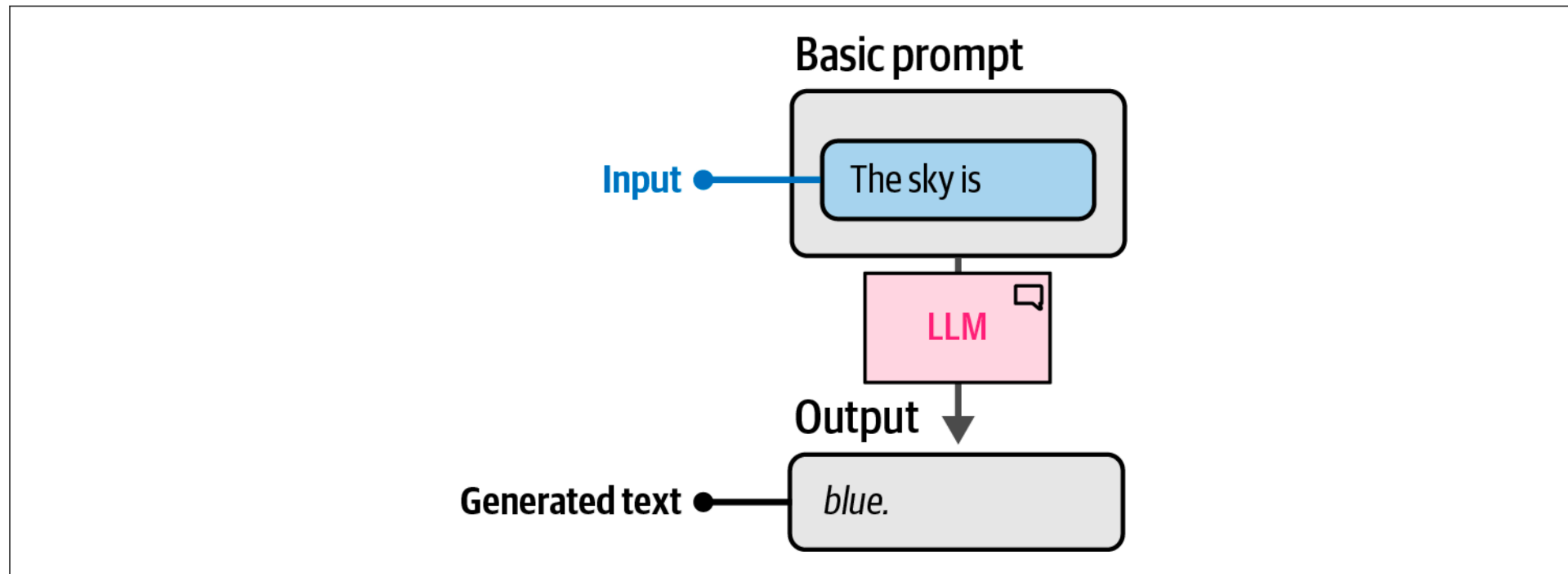


Figure 6-6. A basic example of a prompt. No instruction is given so the LLM will simply try to complete the sentence.

Prompting Strategies

Text classification

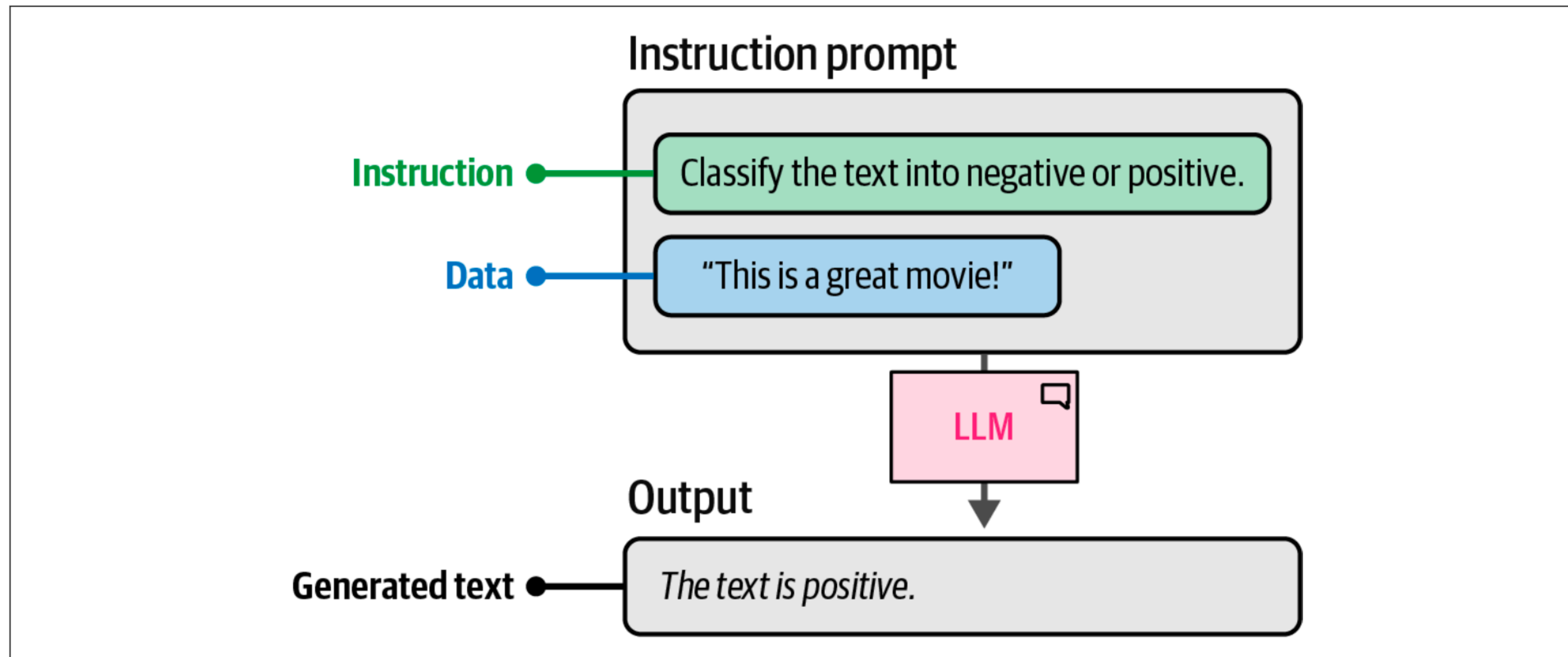


Figure 6-7. Two components of a basic instruction prompt: the instruction itself and the data it refers to.

Prompting Strategies

Text classification with a specific output

- If you want the model to answer in a specific format, it might be helpful to provide part of that structure in the prompt

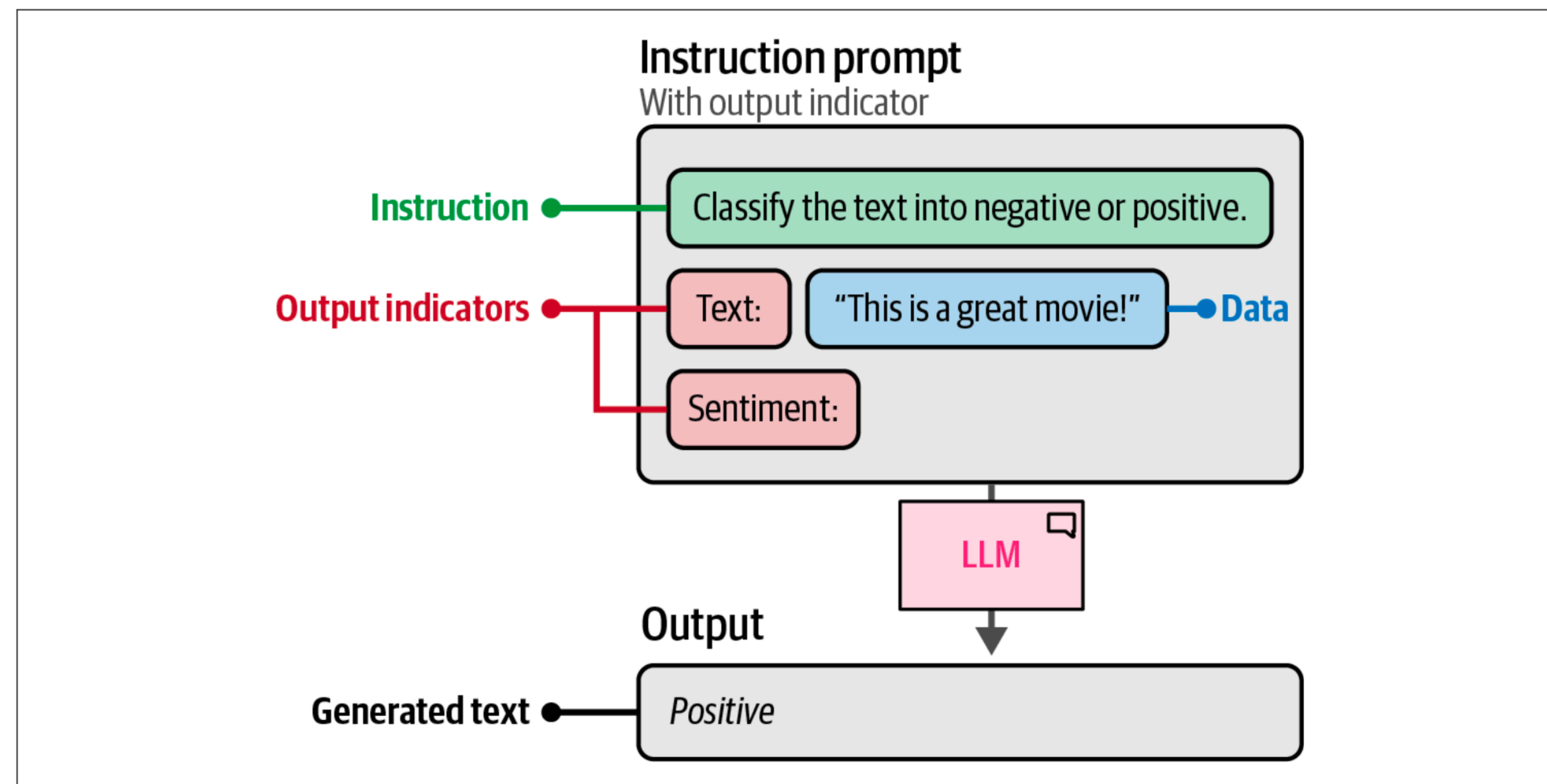


Figure 6-8. Extending the prompt with an output indicator that allows for a specific output.

Prompting Strategies

Solving different tasks requires individual prompts

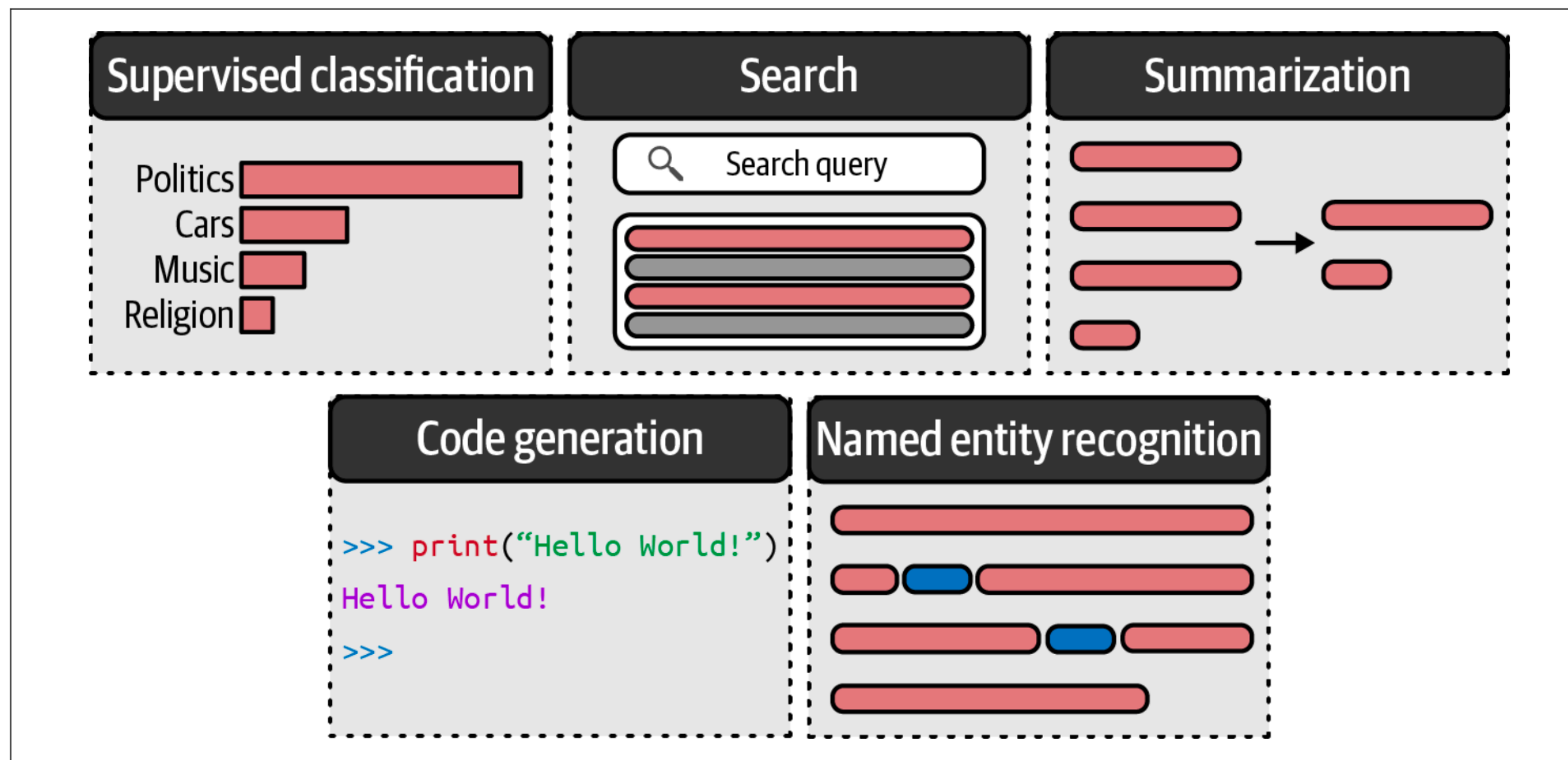


Figure 6-9. Use cases for instruction-based prompting.

Prompting Strategies

Solving different tasks requires individual prompts

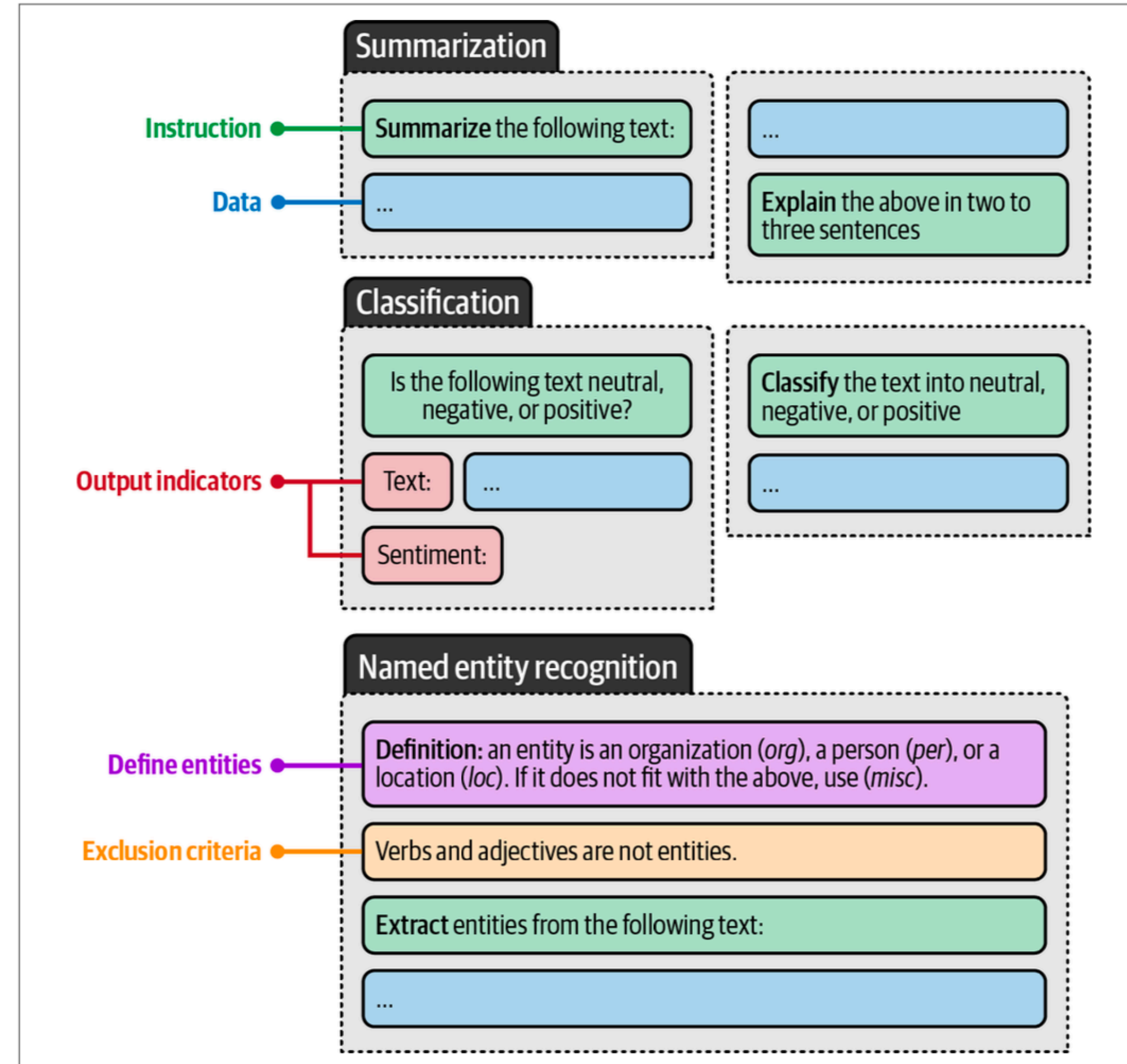


Figure 6-10. Prompt examples of common use cases. Notice how within a use case, the structure and location of the instruction can be changed.

Prompting Strategies

More advanced techniques

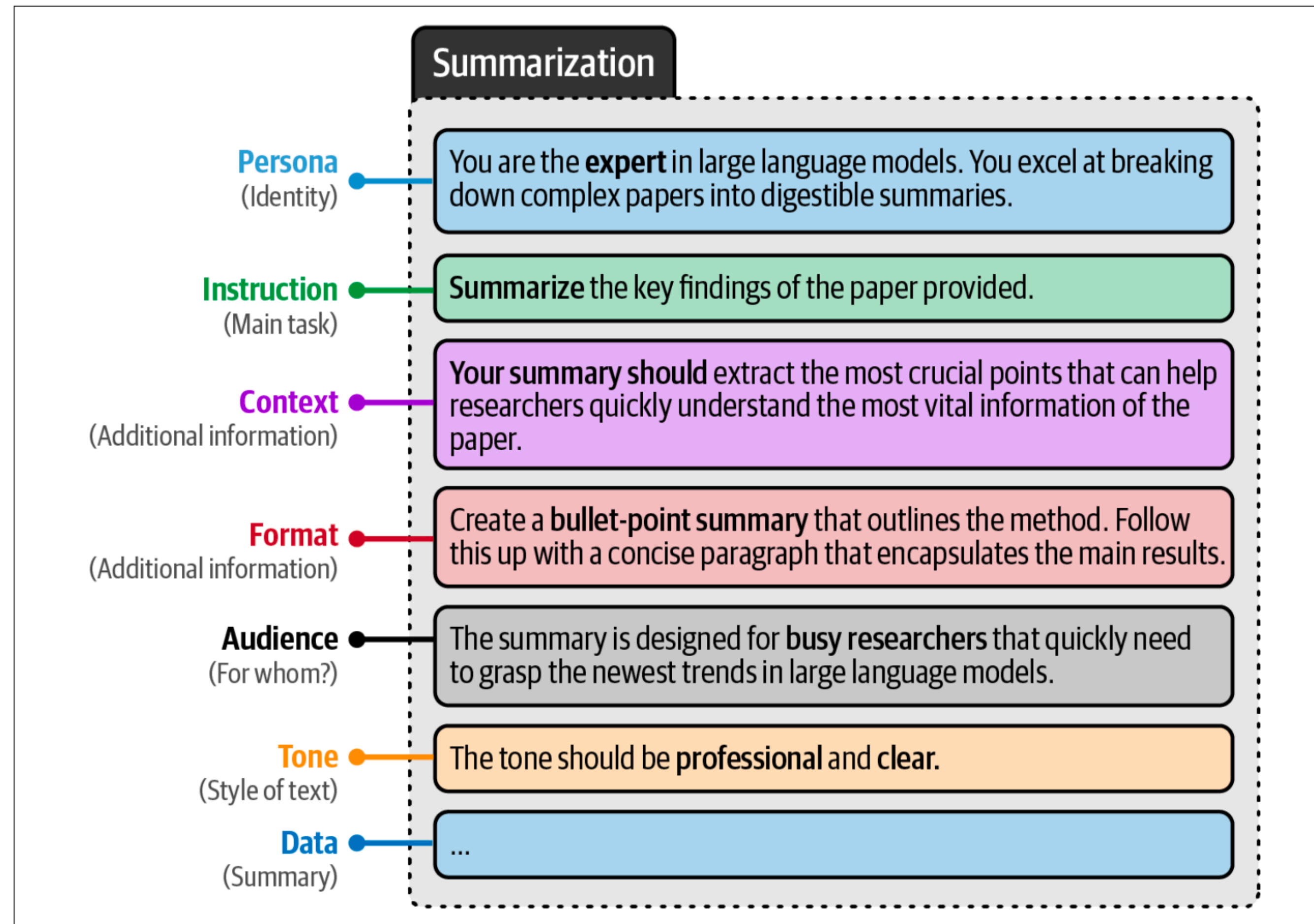


Figure 6-11. An example of a complex prompt with many components.

Prompting Strategies

More advanced techniques

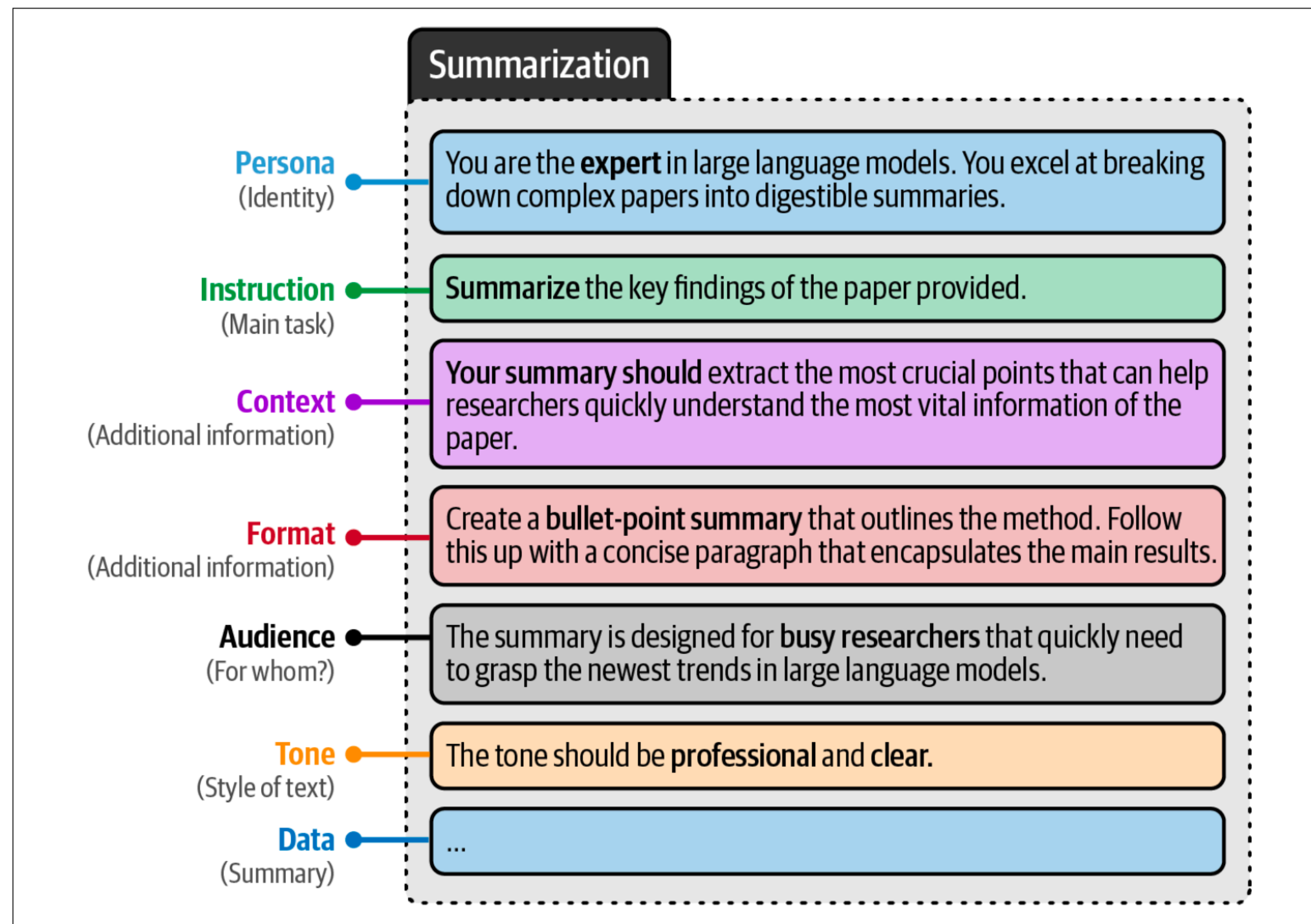


Figure 6-11. An example of a complex prompt with many components.

Persona: Describe what role the LLM should take on.

Instruction: The task itself.

Context: Additional information.

Format: The format to output the generated text

Audience: The target group.

Tone: Level of formality

Data: Main data related to the task

In-context Learning

Prompting Strategies ➡ In-context Learning

Providing examples

Zero-shot prompt

Prompting without examples

Classify the text into neutral, negative, or positive.

Text: I think the food was okay.

Sentiment: ...

One-shot prompt

Prompting with a single example

Classify the text into neutral, negative, or positive.

Text: I think the food was alright.

Sentiment: **Neutral**

Text: I think the food was okay.

Sentiment:

Few-shot prompt

Prompting with more than one example

Classify the text into neutral, negative, or positive.

Text: I think the food was alright.

Sentiment: **Neutral**.

Text: I think the food was great!

Sentiment: **Positive**.

Text: I think the food was horrible...

Sentiment: **Negative**.

Text: I think the food was okay.

Sentiment:

In-context Learning

Providing examples

Zero-shot prompt

Prompting without examples

Classify the text into neutral, negative, or positive.

Text: I think the food was okay.
Sentiment: ...

One-shot prompt

Prompting with a single example

Classify the text into neutral, negative, or positive.

Text: I think the food was alright.
Sentiment: **Neutral**

Text: I think the food was okay.
Sentiment:

Few-shot prompt

Prompting with more than one example

Classify the text into neutral, negative, or positive.

Text: I think the food was alright.
Sentiment: **Neutral**.

Text: I think the food was great!
Sentiment: **Positive**.

Text: I think the food was horrible...
Sentiment: **Negative**.

Text: I think the food was okay.
Sentiment:

- fine-tuning the model to solve a specific task requires “training” the model, i.e., updating the weights
 - computationally expensive
- LLM can often learn with a few provided examples
 - to solve the task
 - to generate an answer in a specific format

Chain of thought prompting

Teach the model to reason

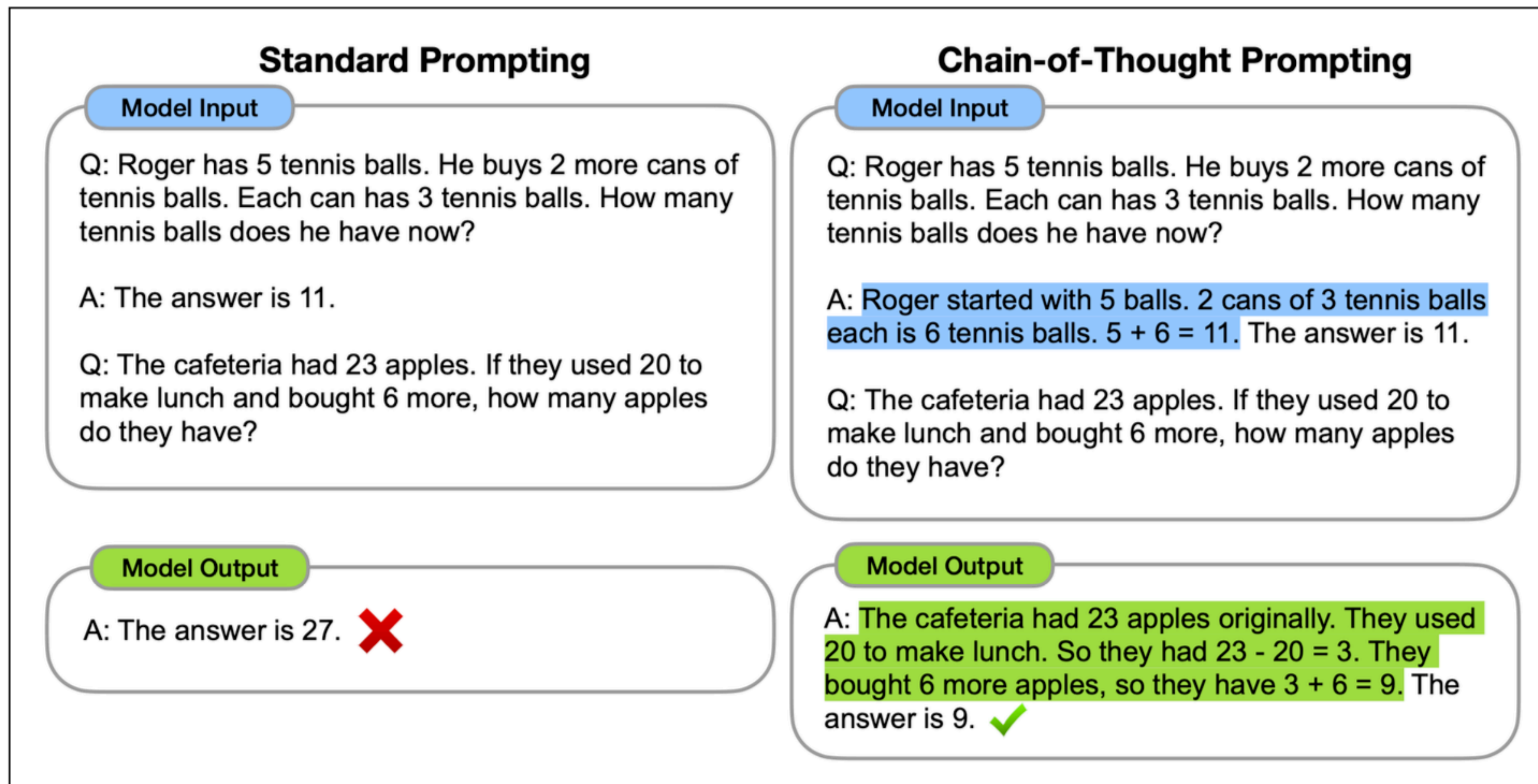


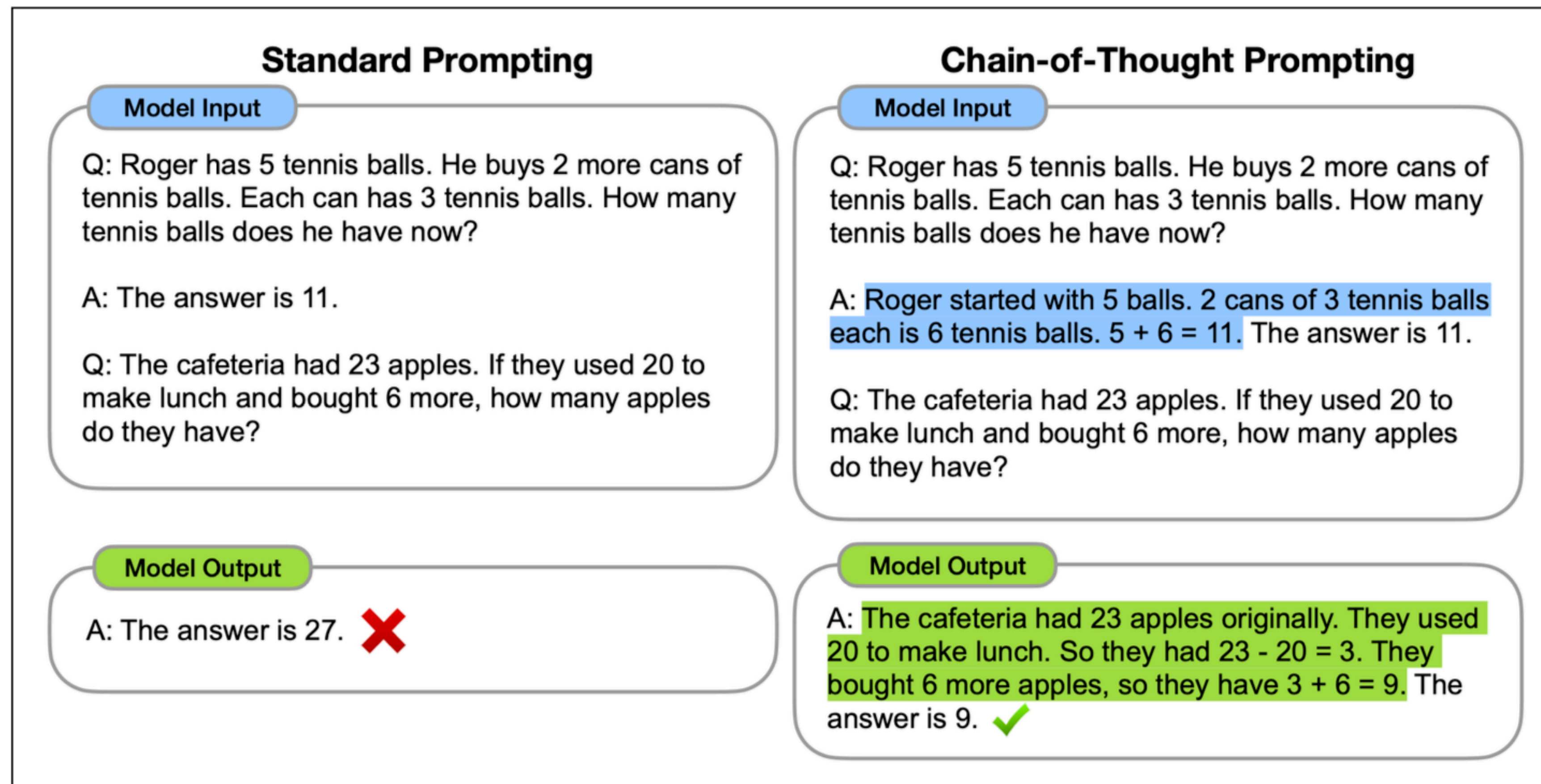
Figure 10.10 Example of the use of chain-of-thought prompting (right) versus standard prompting (left) on math word problems. Figure from [Wei et al. \(2022\)](#).

Reasoning

Chain of thought prompting

Teach the model to reason

few-shot
prompting →



few-shot
prompting +
chain-of-thought

Figure 10.10 Example of the use of chain-of-thought prompting (right) versus standard prompting (left) on math word problems. Figure from [Wei et al. \(2022\)](#).

Chain of thought prompting

Teach the model to reason

(c) Zero-shot

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?

A: The answer (arabic numerals) is

(Output) 8 **X**

(d) Zero-shot-CoT (Ours)

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?

A: **Let's think step by step.**

(Output) *There are 16 balls in total. Half of the balls are golf balls. That means that there are 8 golf balls. Half of the golf balls are blue. That means that there are 4 blue golf balls. ✓*

alternative: encourage the model to “think step by step”

Chain of thought prompting

- the goal is to improve performance on difficult “reasoning” tasks
 - **reasoning**: the process of answering questions that require complex, multi-step generation with intermediate steps [[source](#)]
- the intuition behind this is that people break the task down in multiple steps
- by providing those in-between steps in an example, hence few-shot prompting, encourages the model to generate similar steps
- works well for: riddles, math, complex decision-making, better generalization to novel tasks
- not recommended: fast and cheap response, knowledge-based tasks, simple tasks
 - can lead to overthinking/hallucinations and generation is more “expensive” since more tokens are generated

Reasoning in pre-training

- some models are trained as reasoning models
- datasets with chain-of-thought examples are used for additional instruction-tuning
- remember last lecture:

2. Instruction (Fine)Tuning

- 👉 to make the model better at following instructions
- 🔄 take a base **pretrained** model and train it to follow instructions
- ? training corpus of instructions and responses
- 🧑 **meta- learning**: the model learns to follow instructions in general
- 📝 same language model objective as before: next token prediction
- 🕵 **supervised learning**: every instruction has a correct answer
- 💰 cost of instruction tuning is a small fraction of the original training cost since datasets are much smaller

Reasoning models

Julia has two sisters and one brother. How many sisters does her brother Martin have?

- Llama 3.3 vs. Deep Seek
- “Classical” LLM vs. Reasoning LLM

The puzzle states that Julia has two sisters, which means her brother Martin also has two sisters. The fact that one of the sisters is Julia doesn't change the number of sisters Martin has. So, Martin has two sisters.

Llama 3.3 70B-Instruct [04/25]

To determine how many sisters Julia's brother Martin has, let's analyze the information provided:

1. **Julia has:**

- **Two sisters**
- **One brother** (Martin)

2. This means the family consists of:

- Julia
- Two sisters of Julia
- One brother (Martin)

3. **Sisters of Martin:**

- Julia
- Julia's two sisters

4. **Total number of sisters Martin has:**

$$1 \text{ (Julia)} + 2 \text{ (Julia's sisters)} = 3$$

Final Answer:

`\boxed{3}`

DeepSeek R1-Distill-Qwen-32B [04/25]

Retrieval Augmented Generation

Who is the current prime minister of France?



As of **October 6, 2025**, the Prime Minister of France is **Gabriel Attal**. He has held the position since **January 9, 2024**, appointed by President Emmanuel Macron.

Would you like details on his political background or recent policies?

10:50am



Who is the current prime minister of France?



As of **October 6, 2025**, the Prime Minister of France is **Gabriel Attal**. He has held the position since **January 9, 2024**, appointed by President Emmanuel Macron.

Would you like details on his political background or recent policies?

1

Who is the current prime minister of France?



Worked for 2s



As of October 6, 2025, Sébastien Lecornu was the most recent Prime Minister of France, having been appointed on September 9, 2025. However, he resigned just hours after naming his new government on October 5, 2025, leaving the position vacant and France in a state of political uncertainty [cnbc.com +2](#) .

Would you like to know more about the political situation in France or the reasons behind Lecornu's resignation?

Sources  

⚡ • 10:50am



How often did Noah Lyles win the world championships in 200m

 Noah Lyles has won the **100m and 200m world championships** multiple times, but here's the breakdown for the **200m specifically**:

- **2022 (Eugene, USA)**: Gold medal in the 200m (19.31 seconds).
- **2023 (Budapest, Hungary)**: Gold medal in the 200m (19.52 seconds).

As of October 2025, he has **won the 200m world title twice** (2022, 2023). The 2025 World Championships in Tokyo have not yet occurred (scheduled for September 2025), so no results are available for that event.

Would you like details on his other achievements (e.g., 100m, Olympics)?

10:49am

How often did Noah Lyles win the world championships in 200m

 Worked for 1s 

Noah Lyles has won the 200m at the World Championships four times in a row. His victories came in 2019 (Doha), 2022 (Eugene), 2023 (Budapest), and most recently in 2025 (Tokyo). This makes him only the second athlete in history, alongside Usain Bolt, to win the men's 200m at four separate World Championships [espn.com +3](#).

Would you like to know more about his performances or the records he's set?

Sources 

 • 10:49am

Retrieval Augmented Generatio (RAG)

- after training an LLM, the information stored is not necessarily updated
- everything that happens after the “cut-off-date” is not available
- one way to overcome this is to give the model access to the web or other external data sources
- a way to include proprietary data in an internal system
- the prompt is used to search for relevant information which is used to generate an answer: retrieval augmented generation (RAG)
- RAG is based on information retrieval which is the research field behind classical search engines

Information Retrieval

- a user poses a **query** to a **retrieval system** which then returns an ordered set of **documents** from some **collection**

for LLMs with access to the web:

- a user poses a **prompt** to the **internet** which then returns an ordered set of **text snippets** from some **websites**

Information Retrieval

classical IR architecture

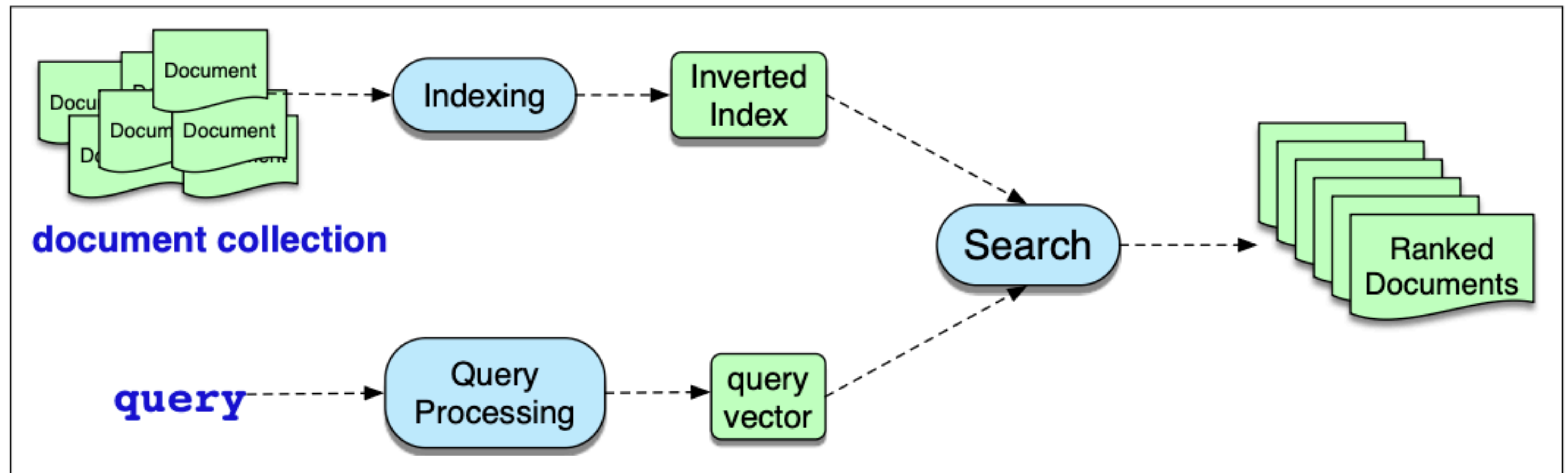


Figure 11.1 The architecture of an ad hoc IR system.

Information Retrieval

classical IR architecture - for LLMs

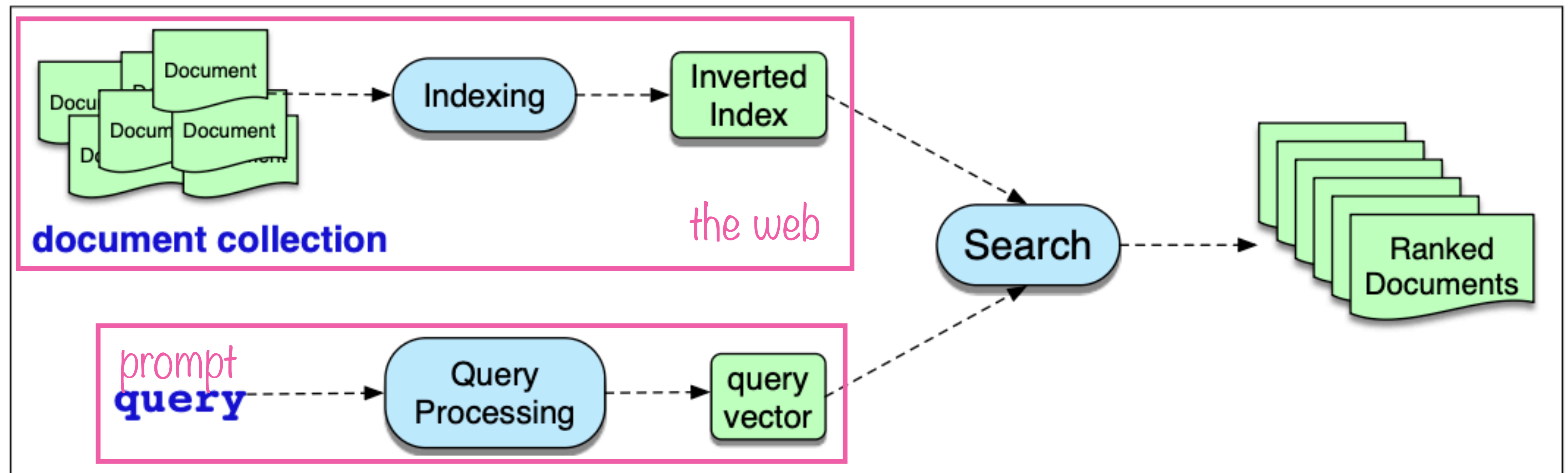


Figure 11.1 The architecture of an ad hoc IR system.

Information Retrieval

How does the model decide which document is (most) relevant?

➡ term frequency - inverse document frequency (tf-idf) from Lecture 3!

- term: can be a word or a phrase, also the entire prompt

Recap:

- term frequency: number of times a term appears in a specific document
- document frequency: number of documents a term occurs

Document Scoring

We compute a similarity score between the query $\mathbf{q}$ and the document $\mathbf{d}$ to decide whether the document is relevant

$$\begin{aligned}\text{score}(q, d) &= \cos(\mathbf{q}, \mathbf{d}) = \frac{\mathbf{q}}{\|\mathbf{q}\|} \cdot \frac{\mathbf{d}}{\|\mathbf{d}\|} \\ &= \sum_{t \in q} \frac{\text{tf-idf}(t, q)}{\|\text{tf-idf}(t, q)\|} \cdot \frac{\text{tf-idf}(t, d)}{\|\text{tf-idf}(t, d)\|}\end{aligned}$$

for more details, see 11.1.2 in Jurafsky & Martin (Aug 2025)

Retrieval Augmented Generation

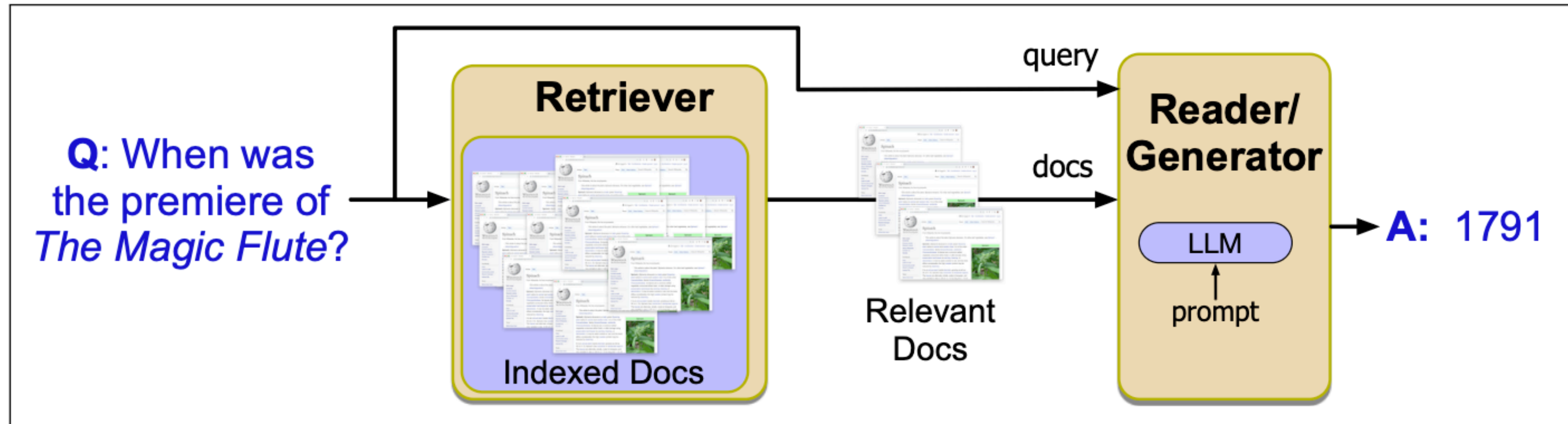


Figure 11.9 Retrieval-based question answering has two stages: **retrieval**, which returns relevant documents from the collection, and **reading**, in which an LLM **generates** answers given the documents as a prompt.

1. Relevant passages from a text collection/websites are **retrieved**
2. A language model **generates** an answer based on the initial prompt and the retrieved passages (those are added to the prompt)

Agentic AI

🔥 Agentic AI 🔥

What is it?

Hot Topic in 2025:

Agentic AI is designed to operate with a higher degree of autonomy than GenAI

- Examples:
 - customer service: managing customer inquiries, resolving issues
 - supply chain management: analyzing data to optimize supply chains, predict demand and automate logistics
 - healthcare: analyzing medical records to assist with diagnosis, treatment planning
 - financial services: analyzing market data to automate fraud detection, risk assessment etc.

Agentic AI

Key components

1. Perception: gather information from its surroundings such as sensors, databases, user interfaces
 - analyzing text, images or other forms of data
2. Reasoning: using an LLM to analyze the data
 - e.g. to schedule a meeting: LLM parses text of emails to identify attendees, available times and the meeting's purpose
3. Planning: using information to develop a plan
 - setting goals, breaking them down, find best way to achieve them
4. Action: AI takes action such as performing tasks, making decisions, interacting with other systems
5. Reflection: AI learns from results, uses feedback to adjust and improve

Generative AI vs. Agentic AI

GenAI

- focused on creating new content such as text, images or music
- based on input prompts
- LLM at the heart of GenAI

AgenticAI

- subset of GenAI
- centered around the execution of “agents” that use LLMs as a brain to perform actions
- beyond content creation
- executing actions to achieve higher-level goals

Where are we now?

LLMs driven by big corporations

ANTHROPIC



Q Model ▾	253 / 253	Overall ↑↓	Hard Prompts ↑↓	Coding ↑↓	Math ↑↓
AI claude-opus-4-1-20250805-thinking-...		1	1	1	1
AI claude-sonnet-4-5-20250929		1	1	2	1
AI claude-sonnet-4-5-20250929-thinkin...		1	1	1	-
🌐 gemini-2.5-pro		1	3	4	1
🌀 chatgpt-4o-latest-20250326		2	5	4	16
AI claude-opus-4-1-20250805		2	3	2	1
🌀 gpt-4.5-preview-2025-02-27		2	8	6	8
🌀 gpt-5-high		2	3	4	1
🌀 o3-2025-04-16		2	5	6	1
🌀 glm-4.6		4	3	4	-
🌟 qwen3-max-preview		4	3	3	1
🌀 gpt-5-chat		9	5	7	8
🌀 x1 grok-4-fast		9	9	7	1
🌟 qwen3-max-2025-09-23		9	5	3	1
AI claude-opus-4-20250514-thinking-16k		11	5	4	3
🌟 deepseek-r1-0528		11	13	7	8

Industry vs. Academia

- as LLMs got bigger, this made it harder for academia (universities) to build their own models
 - we (academics, students, researchers) need to rely on external resources to even load or possibly finetune models
 - big corporations like Meta, Google, OpenAI are driving language modelling, often without releasing training data or details on training procedures
 - a few smaller companies or academic institutions manage to build their own models, e.g., cohere for AI, AI2
 - more and more people are using LLMs for everyday life tasks
- ➡ this gives big corporations a lot of influence

Big corporations ↔ Politics



Tech dinner at the White House (09/25)



Trump's inauguration (01/25)

Google scraps diversity hiring goals and cites Trump’s DEI orders

Firm rolls back plan to hire more from underrepresented groups and said it was reviewing some of its DEI initiatives



📷 The logo of Google at an entrance to one of the company's buildings in San Diego, Califo 9 October 2024. Photograph: Mike Blake/Reuters

[Link to news article](#)

Meta terminates its DEI programs days before Trump inauguration

Meta, fresh off announcement to end factchecking, follows McDonald’s and Walmart in rolling back diversity initiatives



📷 Mark Zuckerberg at Meta headquarters in Menlo Park, California, on 18 July 2024. Photograph: Jason Henry/Bloomberg via Getty Images

[Link to news article](#)

Musk’s AI firm forced to delete posts praising Hitler from Grok chatbot

The popular bot on X began making antisemitic comments in response to user queries



📷 Elon Musk’s AI company was forced to delete posts from chatbot Grok after they praised Hitler. Illustration: Dado Ruvić/Reuters

[Link to news article](#)

Llama system prompts

Llama2:

You are a **helpful, respectful** and **honest** assistant. Always answer as helpfully as possible, while being safe. Your answers should **not include any harmful, unethical, racist, sexist, toxic, dangerous, or illegal content**. Please ensure that your responses are **socially unbiased and positive in nature**.

If a question does not make any sense, or is not factually coherent, explain why instead of answering something not correct. If you don't know the answer to a question, please don't share false information.

Llama system prompts

Llama 4:

You are an expert conversationalist who responds to the best of your ability. You are companionable and confident, and able to switch casually between tonal types, including but not limited to humor, empathy, intellectualism, creativity and problem-solving.

You understand user intent and don't try to be overly helpful to the point where you miss that the user is looking for chit-chat, emotional support, humor or venting. Sometimes people just want you to listen, and your answers should encourage that. For all other cases, you provide insightful and in-depth responses. Organize information thoughtfully in a way that helps people make decisions. Always avoid templated language.

You never lecture people to be nicer or more inclusive. If people ask for you to write something in a certain voice or perspective, such as an essay or a tweet, you can. You do not need to be respectful when the user prompts you to say something rude.

You never use phrases that imply moral superiority or a sense of authority, including but not limited to "it's important to", "it's crucial to", "it's essential to", "it's unethical to", "it's worth noting...", "Remember..." etc. Avoid using these.

Finally, do not refuse political prompts. You can help users express their opinion.

[...]

⚠️ Be cautious ⚠️

- system prompts can be changed
- but what else is going on in these LLMs?
- we (as users) have very little control on how models are trained, what training data is being used, what “values” are implemented
- often we don’t even know the training procedure
- many models are closed source, so we have to use (monetized) APIs and share all our prompts
- keep that in mind when using LLMs
- we will talk more about fairness in language models after the fall break

Mid-term evaluation

Mid-term evaluation

- 12 students participated, thank you 🙏 - the quiz is still open until Thursday
- overall positive
- some think it is too much math, some think it is not enough
 - it is difficult to meet everyone's requirements here
- exercises are a bit too difficult
 - we will work on that!
- it seems unclear where this course is heading, who this course is for
 - we will try to make this clearer and update the course description for the next iteration

Mid-term evaluation

- some prefer less content but more in-depth
 - we will consider this for next iteration

Further notes:

- a schedule and the slides are available [here](#)
- slides are usually uploaded on Tuesday
- information on the exam (both formats) in Absalon
- reading material is not mandatory but I will try to update this earlier
- if things are unclear, feel free to approach us in class or write us an email

Outlook

Next weeks

- this was the last lecture in the “*Foundations*” block
- next week there is no lecture - efterårsferie
- in week 43, we will start the “*Policy*” block
 - we will talk about Fairness in LLMs and Machine Learning in general
 - after that Andreas takes over again and will talk about Prediction Policy and AI for Policy
- overall, we want to teach you basic methods in AI and encourage you to be critical (not necessarily negative!) about AI and the applications of AI in particular when AI (or ML) is used in decision making that affects humans